# Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

## School of Computing

### Department of Computer Science and Engineering

**10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)**

# PROJECT REVIEW : MILESTONE PROJECT – 2

## Date: 17-04-2026

Smart Campus Event Management System

**SDG: SDG 4 – Quality Education**


**Presented By:**

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**Courses Handling Faculty:**

Dr. Hafizur Rahman-Associate Professor

# INTRODUCTION

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- In today's digital era, managing campus events manually leads to inefficiency, data loss, and lack of proper coordination among students and organizers.

- Many institutions rely on scattered information sources, which creates confusion and makes it difficult for students to stay updated about upcoming events.

- There is a growing need for a centralized system that provides structured event management with real-time access and easy participation.
- The **Campus Event Management System** is a full-stack web application designed to allow administrators to create, manage, and monitor events efficiently.
- The system enables students to view available events, register for them, and access event-related details in a user-friendly interface.

- The application uses a database system (H2/MySQL) to maintain structured data for events, users, and registrations.
- The system is developed using **Spring Boot, Java, Thymeleaf, HTML, CSS, and Bootstrap**, ensuring a responsive and dynamic web application.

# PROBLEM STATEMENT

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In today’s fast-paced campus environment, managing events manually is inefficient and often leads to poor communication and missed opportunities for students. Many institutions rely on notice boards or scattered online updates, which makes it difficult for students to stay informed about upcoming events. There is no centralized system to manage event details, registrations, and participant tracking, creating challenges for administrators in organizing and maintaining records. Additionally, the lack of secure authentication can result in unauthorized access and data inconsistencies. Students also do not have a simple platform to explore and register for events easily. Therefore, there is a need for a centralized, secure, and user-friendly web-based Campus Event Management System that allows administrators to manage events efficiently and enables students to view and register for events seamlessly.

# OBJECTIVE(S)

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- To design and develop a responsive web-based **Campus Event Management System** that allows administrators to create and manage events, and enables students to view and register for events easily using modern web technologies.

- To implement secure authentication and CRUD operations for managing events and registrations, ensuring safe access, efficient data handling, and a user-friendly experience through a relational database system.

# PROPOSED SYSTEM

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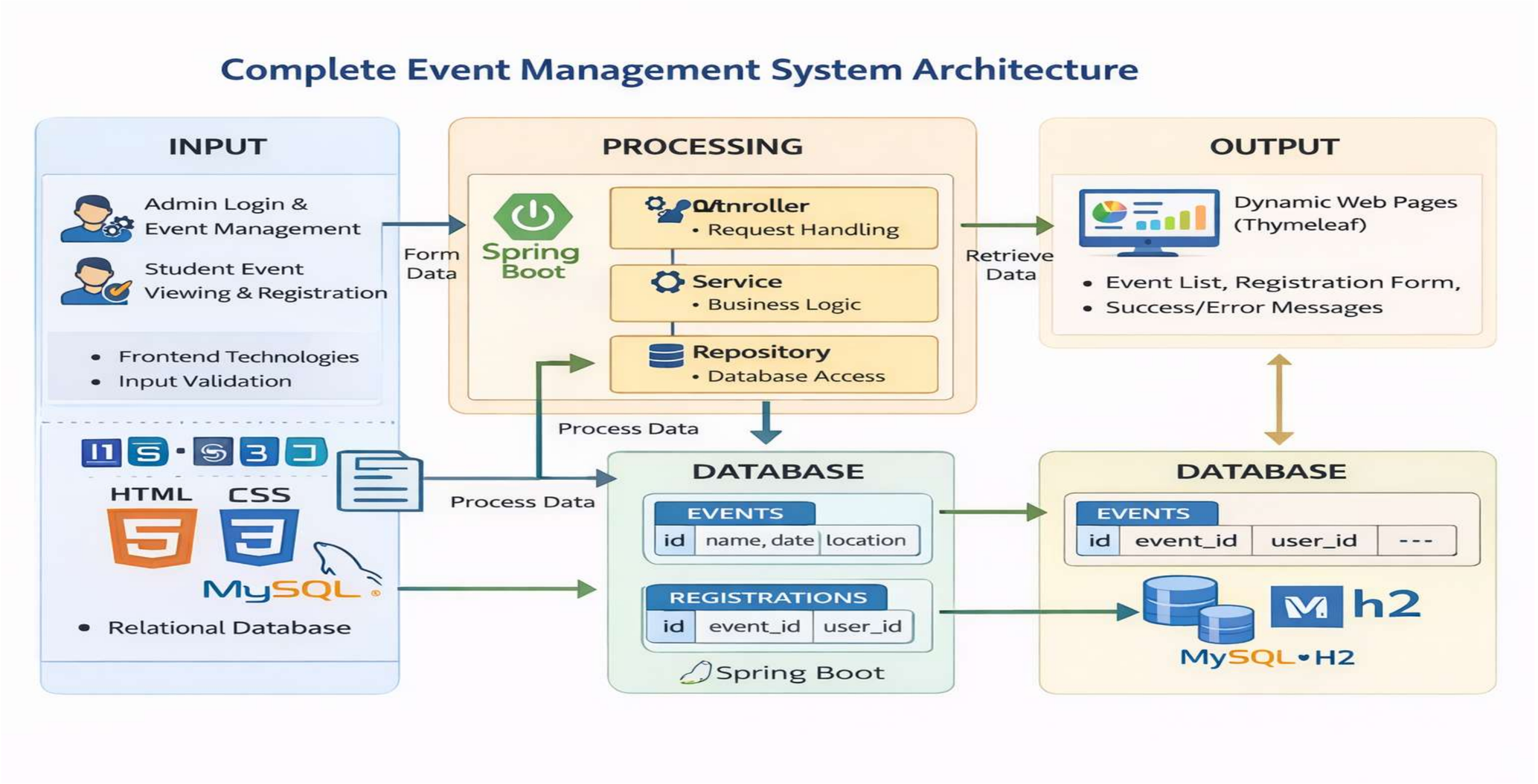
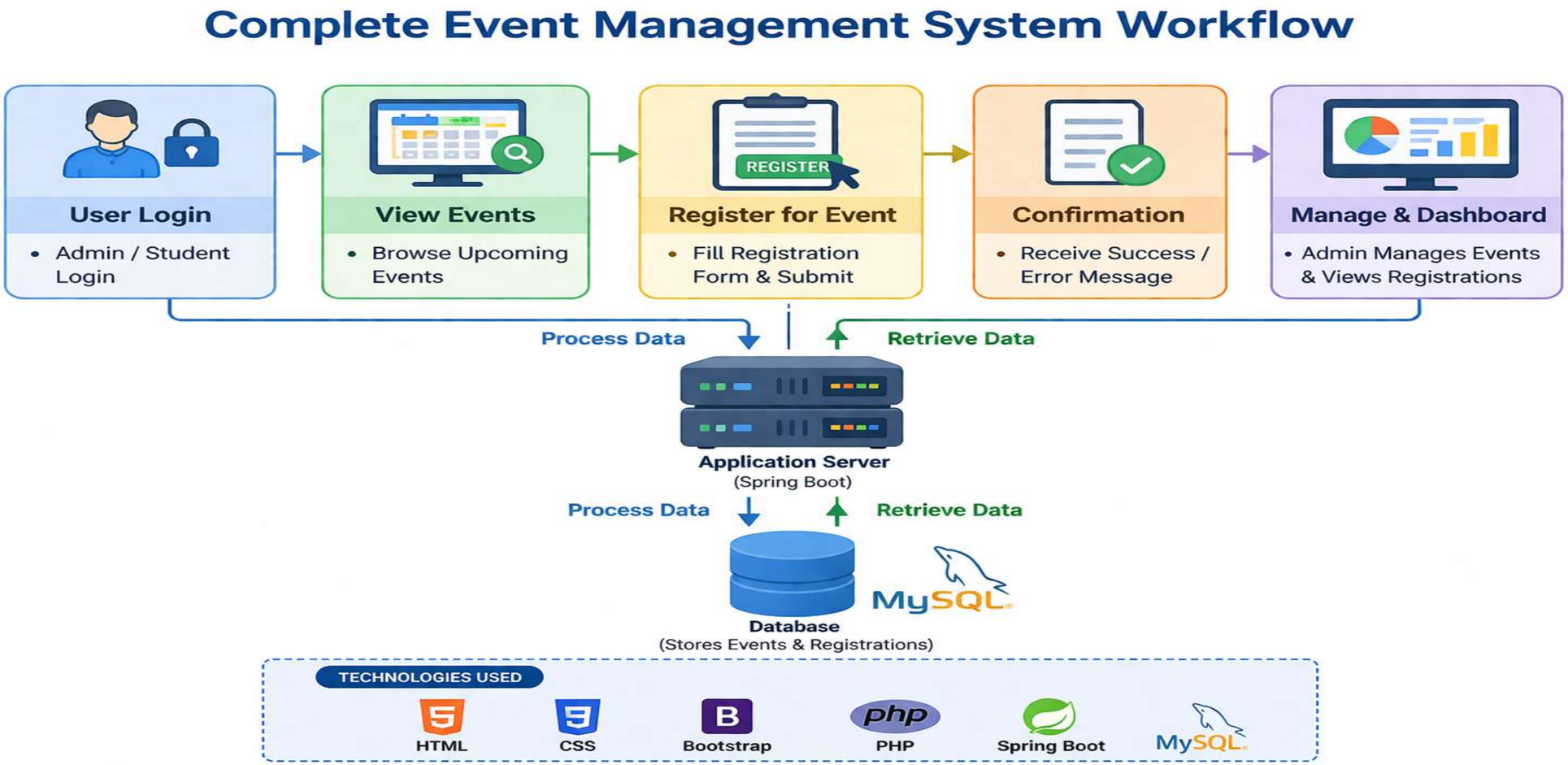
- **Complete  Event Management Solution –**
  The Campus Event Management System is a web-based full-stack application that allows administrators to create, manage, and organize events while enabling students to view and register — providing a complete platform for campus event handling.
- **Secure Authentication –**
  Implements admin login and session management to ensure only authorized users can add, update, or delete events, protecting system data.
- **Dynamic Event Management –**
  Events are stored in a database and displayed dynamically, allowing students to view event details and register in real time.
- **Relational Database Structure –**
  Maintains relationships between Events and Registrations using a structured database for consistency and efficient data management.
- **Student Registration System –**
  Students can easily register for events by providing basic details, and registrations are stored and managed efficiently.
- **How it Solves Existing Problems –**
  - Provides a centralized platform for all campus events
  - Eliminates confusion from scattered information
  - Ensures secure admin control
  - Simplifies event registration process
  - Improves student participation and engagement

- **Key Advantages –**Secure Authentication, Centralized Event System, Easy Registration, Real-time Updates, User-Friendly Interface, Scalable Architecture.

# PROPOSED SYSTEM ARCHITECTURE



# WORKFLOW


Complete Event Management System Workflow
User Login
• Admin / Student Login
View Events
• Browse Upcoming Events
REGISTER
Register for Event
• Fill Registration Form & Submit
Confirmation
• Receive Success / Error Message
Manage & Dashboard
• Admin Manages Events & Views Registrations
Process Data
Retrieve Data
Application Server
(Spring Boot)
Process Data
Retrieve Data
MySQL
Database
(Stores Events & Registrations)
TECHNOLOGIES USED
HTML
CSS
Bootstrap
PHP
Spring Boot
MySQL

# TECHNOLOGIES USED

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- **Frontend:** HTML, CSS, Bootstrap , JavaScript
- **Backend:**Java, Spring Boot, Spring MVC, CRUD Operations, Authentication Logic
- **Database:**MySQL / H2 Database, Events Table, Registrations Table, Relational Mapping (Foreign Key)
- **Tools:**Eclipse / VS Code, Maven, Spring Boot DevTools, Web Browser
- **Hardware Requirements:** Minimum 4GB RAM, 20GB free storage, Intel i3 processor or above
- **Software Requirements:**Windows / Linux / macOS, Java 17 or above, Spring Boot, MySQL Server, Modern Web Browser

# MODULE DESCRIPTION

**Module 1:Admin Authentication & Management**

This module handles admin login and session management within the system. It ensures that only authorized administrators can access the system to add, update, or delete events. Secure authentication mechanisms are used to protect the application and maintain data integrity.

**Module 2: Event Management**

This module manages all event-related operations including adding, viewing, updating, and deleting events. It stores details such as event name, department, type, date, and description. Events are dynamically displayed for students to browse easily.

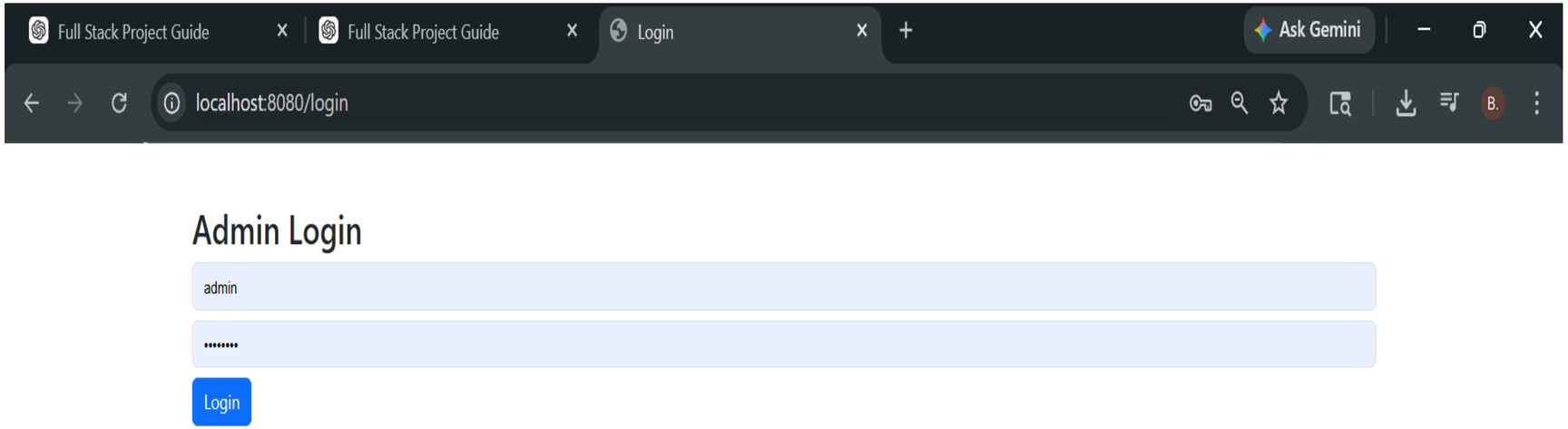
**Module 3:Registration Management**

This module handles the event registration process where students can register for available events. It maintains the relationship between events and participants using a database structure and ensures that each registration is stored and tracked properly.

**Module 4:Event Viewing & Dashboard Module**

This module allows students to view available events and their details through a user-friendly interface. It displays registered events, provides feedback messages, and ensures smooth navigation across the application.

# IMPLEMENTATION SCREENSHOTS

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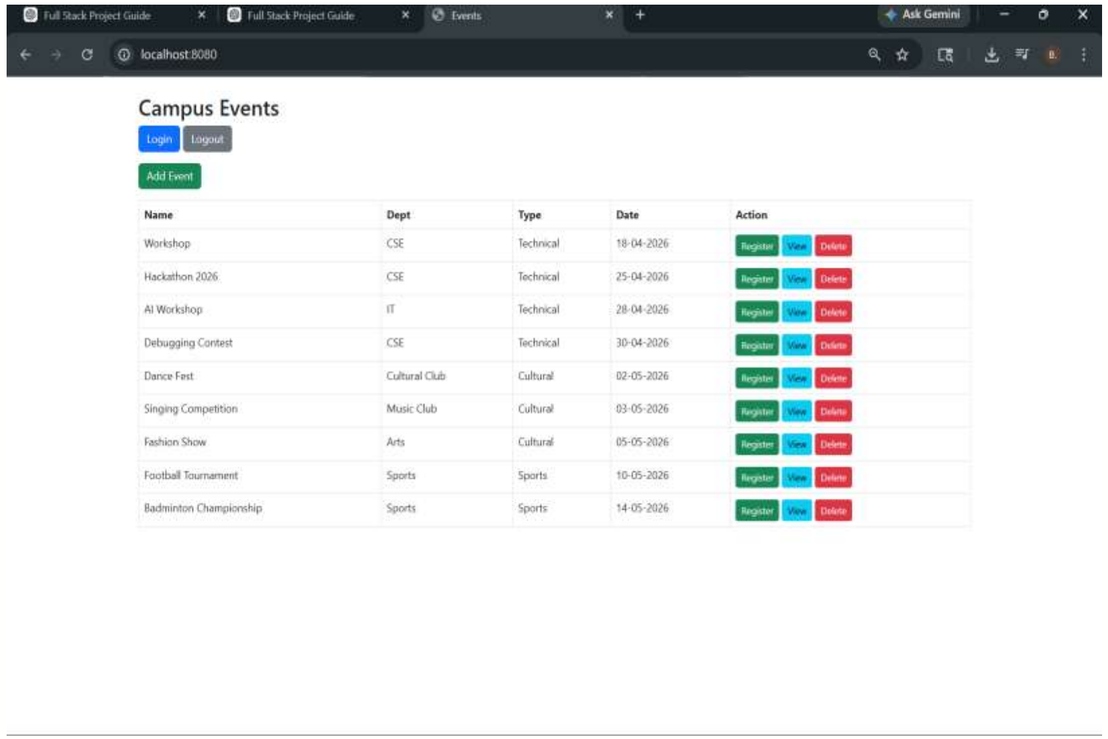


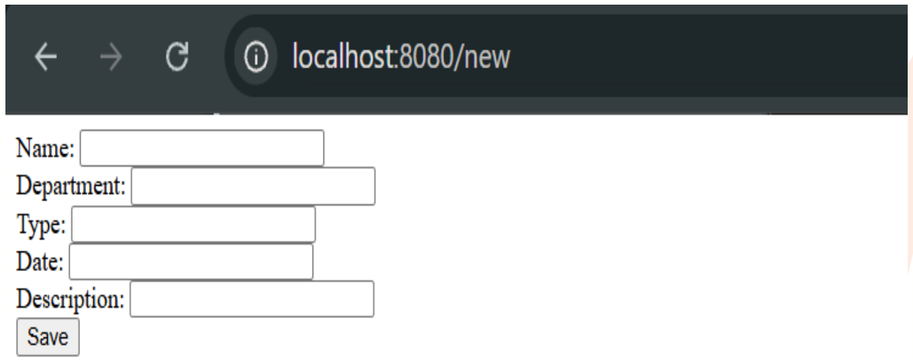


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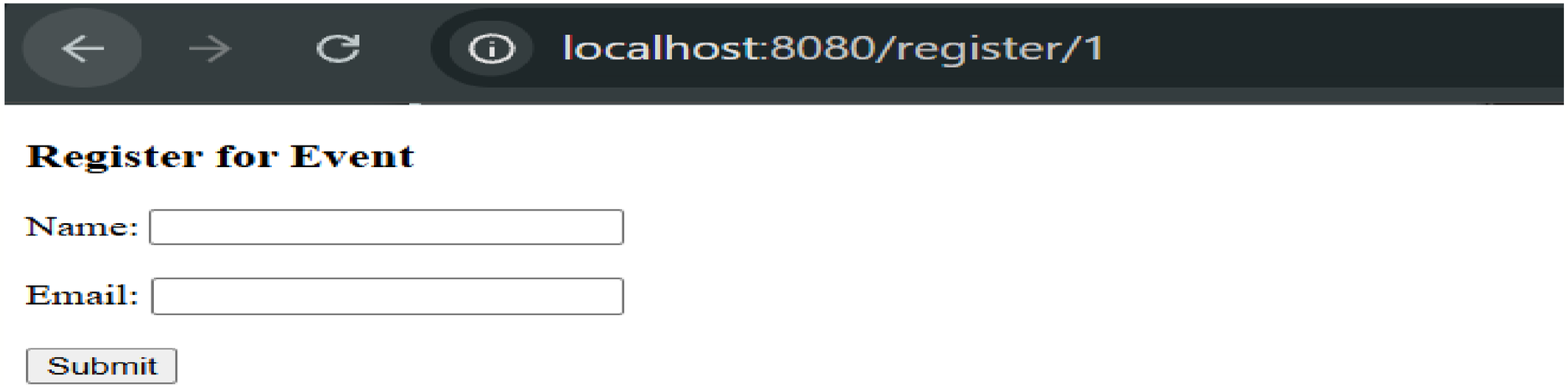


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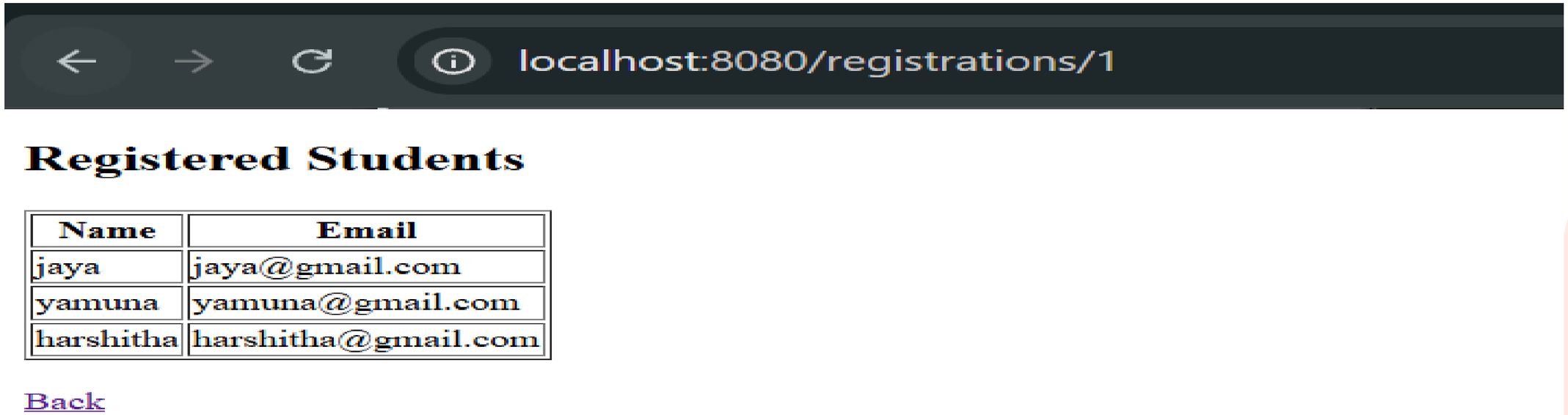


**Event Viewing & Dashboard Module**

This module allows students to view available events and their details through a user-friendly interface. It displays registered events, provides feedback messages, and ensures smooth navigation across the application.

# ADVANTAGES & LIMITATIONS

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**Advantages:**

- Simple and user-friendly interface for easy event browsing and navigation
- Secure admin authentication system for controlled access
- Efficient relational database structure ensuring data integrity
- Supports CRUD operations for managing events and registrations
- Centralized platform for all campus events
- Responsive design compatible with desktop and mobile devices

**Limitations:**

- No payment gateway integration for paid events
- Limited advanced features like notifications or live updates
- Application runs locally and is not deployed on the cloud
- No advanced analytics or reporting dashboard
- Limited multi-user role management (only basic admin access).

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